

INTERNSHIP REPORT

on

THE NEXUS DOME

An IoT-Based Smart Home Automation Ecosystem
Real-Time Monitoring • Automation • Security • Energy
Management



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Embedded System and IoT for Industrial Application

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1. Abstract

The Nexus Dome is an Internet of Things (IoT)-based Smart Home Automation Ecosystem designed to improve home security, automation, energy management, and remote accessibility. The system integrates an ESP32 microcontroller, multiple environmental and security sensors, cloud connectivity, a responsive web dashboard, and a mobile application into a unified platform. It continuously monitors the home environment, automates appliances based on predefined conditions, and provides real-time notifications to users. The architecture has been designed to be scalable, allowing future integration of Artificial Intelligence (AI) features such as behaviour learning, predictive maintenance, and intelligent decision-making.

2. Problem Statement

Modern households continue to face several challenges related to safety, convenience, and energy management. Users often worry about whether appliances have been switched off, doors are securely locked, or whether emergencies such as gas leakage or fire may occur when nobody is at home. Existing smart home solutions generally focus on individual functionalities and are often expensive, making them less accessible for many families.

The need for an affordable, integrated, and remotely accessible smart home solution inspired the development of The Nexus Dome, which combines multiple automation and monitoring features into a single ecosystem.

3. Objectives

The primary objectives of The Nexus Dome are:

1. To develop an affordable IoT-based smart home automation prototype.
2. To provide real-time monitoring of environmental and security parameters.
3. To automate household appliances based on sensor data and predefined thresholds.
4. To improve home safety through gas leakage, fire, motion, earthquake, and unauthorized access detection.
5. To enable secure remote monitoring and control through web and mobile

applications.

6. To monitor household energy consumption using current sensing technology.
 7. To develop a scalable architecture capable of supporting future AI-powered smart home functionalities.
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4. Proposed Solution

The Nexus Dome is designed as a centralized IoT-based smart home ecosystem that combines multiple automation and monitoring functionalities into a single integrated platform. The system uses the ESP32 as the primary controller to collect real-time data from various environmental and security sensors.

Based on predefined threshold values, the ESP32 automatically controls household appliances such as lighting, cooling fans, garage doors, and security mechanisms. At the same time, all sensor readings and device statuses are synchronized with the cloud, allowing users to monitor and control their homes remotely through web and mobile application.

Unlike conventional smart home systems that focus on individual functionalities, The Nexus Dome integrates security, automation, energy monitoring, and remote accessibility within one unified platform while maintaining affordability through open-source hardware and software technologies.

5. System Architecture

The overall architecture of The Nexus Dome consists of four major layers:

1. User Layer

Users interact with the smart home through the responsive web dashboard or the mobile application. Both interfaces provide real-time monitoring, device control, and emergency notifications.

2. Cloud & Backend Layer

The Django backend processes requests from users, stores sensor data in the PostgreSQL database, and manages real-time communication using MQTT and WebSockets.

3. Controller Layer

The ESP32 serves as the central controller, continuously collecting data from sensors, processing automation rules, and controlling actuators such as relays and servo motors.

4. Sensor & Actuator Layer

This layer includes all input and output devices such as the DHT22, MQ2, PIR, Flame Sensor, RFID module, Ultrasonic Sensor, ACS712, MPU6050, Relay Module, Servo Motor, and LCD Display, enabling intelligent monitoring and automation.

6. Hardware Stack

Overview

The hardware architecture of The Nexus Dome is built around the ESP32 microcontroller and a collection of environmental, security, and automation sensors. These components work together to monitor the home environment, detect emergency situations, automate household appliances, and provide real-time data to the cloud platform.

The prototype measures approximately 50 cm × 25 cm and demonstrates the functionality of a complete smart home ecosystem using affordable and easily available electronic components.

Component	Purpose	Operating Range / Specification
ESP32	Central controller responsible for sensor processing, Wi-Fi communication, automation logic, and cloud connectivity.	Dual-Core 240 MHz, Wi-Fi & Bluetooth, 3.3V

DHT22	Measures indoor temperature and humidity for climate monitoring.	Temperature: -40°C to 80°C, Humidity: 0–100% RH
MQ2 Gas Sensor	Detects combustible gases such as LPG, methane, propane, and smoke.	Approximately 300–10,000 ppm
Flame Sensor	Detects fire by sensing infrared radiation emitted from flames.	Detection Distance: 20–100 cm
PIR Motion Sensor	Detects human movement for security and automatic lighting.	3–7 m, 110° viewing angle
4×4 Matrix Keypad	Provides secure PIN-based authentication for the main entrance.	4×4 Matrix Keypad (16 Keys), Operating Voltage: 3.3V–5V
HC-SR04 Ultrasonic Sensor	Measures vehicle distance for automatic garage door control.	2–400 cm, ±3 mm accuracy
Servo Motor	Operates the garage door and main gate locking mechanism.	0°–180° rotation
Relay Module	Controls electrical appliances such as lights and cooling fans.	5V relay module
ACS712 Current Sensor	Measures electrical current consumed by connected appliances.	5A module
MPU6050 Accelerometer	Detects abnormal vibrations for earthquake monitoring.	±2g, ±4g, ±8g, ±16g

**16×2 LCD
Display**

**Displays important system
information locally.**

**16 Characters × 2
Lines**

Sensor Threshold Values

The ESP32 continuously compares sensor readings against predefined threshold values. Whenever a sensor crosses its threshold, the system automatically performs the corresponding action.

Sensor	Threshold (Prototype)	System Response
DHT22	Temperature > 30°C	Cooling fan turns ON
MQ2	Gas concentration > 500 ppm	Alarm activated and emergency notification sent
Flame Sensor	Flame detected within sensing range	Buzzer activated and emergency alert generated
PIR Sensor	Motion detected	Lights turn ON (during low-light conditions)
Keypad	Valid PIN Entered	Main gate unlocks
Ultrasonic Sensor	Vehicle detected within 15 cm	Garage door opens automatically
ACS712	Current > 3 A	High power usage warning displayed
MPU6050	Sudden acceleration > 1.5 g	Earthquake alert generated

7. Software Stack

Overview

The software architecture of **The Nexus Dome** combines embedded programming, backend development, database management, frontend technologies, cloud deployment, and version control to create a complete IoT-based smart home automation ecosystem. Each technology was selected to perform a specific role, ensuring reliable communication between the ESP32, cloud services, web dashboard, and mobile application.

Software Technologies Used

Technology	Purpose in The Nexus Dome
Wokwi	Used to simulate and test the ESP32 circuit, sensors, and automation logic before implementing the physical prototype.
Python	Primary programming language used for backend development, sensor data processing, and system logic.
Django	Backend framework responsible for user authentication, device management, APIs, and application logic.
Django REST Framework	Used to build secure REST APIs for communication between the frontend, mobile application, and backend server.
PostgreSQL	Stores user accounts, device information, sensor readings, automation settings, and historical data.
Scikit-learn	Integrated for future-ready machine learning features and predictive analytics. The architecture supports AI expansion in upcoming versions.
React.js	Used to develop the responsive web dashboard for real-time monitoring and remote control.
React Native	Used to build the Android mobile application for controlling the smart home from anywhere.

Firebase	Provides push notifications and cloud messaging services for emergency alerts and important system events.
Railway	Cloud platform used to deploy and host the Django backend and PostgreSQL database.
Vercel	Hosts and deploys the React web dashboard, providing fast and reliable frontend access.
GitHub	Used for version control, project collaboration, and source code management.

Software Development Workflow

The software development was carried out in a modular manner, allowing each component to be tested independently before complete system integration.

Step 1 – Prototype Simulation

The hardware connections and automation logic were first verified using **Wokwi**, reducing hardware debugging time before implementation on the physical prototype.

Step 2 – Backend Development

The backend was developed using **Python**, **Django**, and **Django REST Framework**, providing secure APIs, authentication, and communication between users and IoT devices.

Step 3 – Database Design

A **PostgreSQL** database was created to manage user profiles, registered devices, sensor readings, and automation configurations.

Step 4 – Frontend Development

A responsive **React.js** web dashboard and a **React Native** mobile application were developed for real-time monitoring and remote control of the smart home.

Step 5 – Cloud Deployment

The Django backend and PostgreSQL database were deployed on **Railway**, while the React web application was deployed on **Vercel**. **Firebase** was integrated to deliver instant notifications during emergency situations.

Step 6 – Version Control

Throughout development, **GitHub** was used to manage source code, track project progress, and enable collaborative development among team members.

Key Software Features

1. Secure user authentication and device management.
 2. Responsive web dashboard and Android mobile application.
 3. Real-time monitoring of sensor data.
 4. Remote control of appliances from anywhere.
 5. Cloud-based backend and database deployment.
 6. Push notifications for emergency alerts.
 7. Modular architecture supporting future AI integration.
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8. Working of The Nexus Dome

Overview

The **Nexus Dome** operates as an integrated IoT-based smart home automation system where the ESP32 acts as the central controller. It continuously collects data from multiple sensors, processes the information using predefined automation rules, and performs appropriate actions. Simultaneously, the system synchronizes all sensor readings and device statuses with the cloud, allowing users to monitor and control their homes remotely through the web dashboard and mobile application.

The working of the system can be divided into the following stages:

Step 1 – Data Acquisition

Environmental and security sensors continuously monitor different parameters inside the house.

The sensors used in the prototype include:

- DHT22 for temperature and humidity
- MQ2 for gas leakage detection
- Flame Sensor for fire detection
- PIR Sensor for motion detection
- 4×4 Matrix Keypad for secure PIN-based access
- HC-SR04 Ultrasonic Sensor for garage automation

- ACS712 for current monitoring
- MPU6050 Accelerometer for earthquake detection

The ESP32 reads these sensor values at regular intervals.

Step 2 – Data Processing

The ESP32 compares the incoming sensor values with predefined threshold values stored in the program.

Examples include:

- Temperature greater than **30°C**
- Gas concentration above **500 ppm**
- Flame detected
- Vehicle detected within **15 cm**
- Current exceeding **3 A**
- Sudden vibration indicating an earthquake

Based on these conditions, the ESP32 decides whether any action is required.

Step 3 – Intelligent Automation

Once a condition is satisfied, the ESP32 automatically controls the corresponding devices.

Examples include:

- Switching ON the cooling fan during high temperature.
- Activating the alarm when gas leakage or fire is detected.
- Unlocking the main gate after successful PIN verification through the keypad.
- Opening the garage door automatically when a vehicle approaches.
- Turning ON room lights when motion is detected under low-light conditions.
- Generating an emergency alert during abnormal vibrations indicating a possible earthquake.

This rule-based automation reduces manual intervention while improving safety and convenience.

Step 4 – Cloud Synchronization

After processing the sensor data, the ESP32 sends the latest information to the backend server.

The backend stores:

- Sensor readings
- Device status
- User information
- Automation logs
- Energy consumption history

This enables users to access live information from anywhere.

Step 5 – Remote Monitoring & Control

Users can access **The Nexus Dome** using either the web dashboard or the Android mobile application.

The interface allows users to:

- Monitor live sensor readings.
- View security status.
- Check environmental conditions.
- Monitor electricity consumption.
- Control appliances remotely.
- Receive emergency notifications.

All updates are synchronized in real time, providing a seamless user experience.

Step 6 – Continuous Monitoring

The entire process repeats continuously, ensuring uninterrupted monitoring of the smart home.

The ESP32 constantly collects sensor data, updates the cloud database, performs automation, and responds immediately whenever abnormal conditions are detected.

9. Web Dashboard & Mobile Application

Overview

To provide users with complete control over their smart home, **The Nexus Dome** includes a responsive **web dashboard** and an **Android mobile application**. Both platforms communicate with the backend server and allow users to monitor sensor data, control appliances, receive emergency alerts, and manage home security from anywhere with an internet connection.

The dashboard has been designed with a simple and intuitive interface, making it suitable for users of all age groups while providing real-time information about the entire smart home ecosystem.

Dashboard Modules

The web dashboard is divided into multiple functional modules, each responsible for monitoring and controlling a specific aspect of the smart home.

1. Dashboard

The main dashboard provides a complete overview of the house by displaying:

- Live temperature and humidity
- Gas concentration
- Current consumption
- Motion detection status
- Garage status
- Flame detection status
- Quick controls for lighting and cooling fan
- Recent temperature trend

This enables users to monitor the overall condition of their home from a single screen.

2. Security Module

The Security page focuses on access control and home protection.

It includes:

- Keypad-based door lock access
- Garage door automation
- Motion monitoring
- Secure access status
- Manual gate control

The garage door automatically opens when the ultrasonic sensor detects an approaching vehicle and closes once the vehicle has entered.

3. Climate Module

The Climate page allows users to monitor indoor environmental conditions.

It displays:

- Temperature
- Humidity
- Thermostat settings
- Fan control

When the temperature exceeds the predefined threshold, the cooling fan is activated automatically.

4. Safety & Alarm Module

The Safety page continuously monitors emergency conditions inside the house.

It provides information regarding:

- Gas leakage
- Fire detection
- Earthquake detection using MPU6050
- Current overload
- Emergency alarm status

Whenever an abnormal condition is detected, the system immediately activates the buzzer and sends a notification to the user.

5. Energy Module

The Energy dashboard helps users understand their electricity usage.

Features include:

- Real-time current monitoring
- Estimated power consumption
- Energy usage trends
- Appliance control

This enables users to identify excessive power consumption and improve energy efficiency.

6. Household Module

The Household page manages registered users and device ownership.

Users can:

- View authorized family members
- Manage household access

- Monitor connected smart devices

This ensures that only authorized users can control the smart home.

7. Profile Module

The Profile page displays user and home information.

It includes:

- User profile
- Home details
- Connected devices
- Account status

This page allows users to manage their personal account information.

Mobile Application

In addition to the web dashboard, **The Nexus Dome** also provides an Android mobile application developed using **React Native**.

The mobile application allows users to:

- Monitor sensor readings remotely.
- Receive emergency notifications.
- Control household appliances.
- Check home security status.
- Access the smart home from anywhere.

The mobile interface provides the same functionality as the web dashboard while offering greater convenience for daily use.

Key Advantages

1. User-friendly interface
 2. Responsive design
 3. Remote access through internet
 4. Real-time monitoring
 5. Simple navigation
 6. Secure user authentication
 7. Easy appliance control
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10. Innovation & Business Vision

Innovation

The Nexus Dome is not just a collection of smart devices; it is a unified IoT ecosystem that integrates security, automation, climate monitoring, energy management, and remote accessibility into a single platform. Instead of using multiple independent applications for different smart devices, users can monitor and control all connected devices through one centralized web dashboard and mobile application.

The prototype has been designed using affordable hardware components while maintaining a modular architecture that can be easily expanded in the future. This makes the system suitable for both educational purposes and practical smart home applications.

Key Innovations

- **Centralized Smart Home Platform** integrating multiple functionalities into one ecosystem.
- **Affordable Prototype** developed with a total hardware cost of approximately ₹3,000, making smart home technology more accessible.
- **Real-Time Monitoring** of environmental, security, and energy parameters.
- **Automatic Rule-Based Automation** for lighting, cooling, garage access, and emergency response.
- **Responsive Web Dashboard** for complete remote monitoring and device control.
- **Android Mobile Application** providing users with access to their smart home from anywhere.
- **Scalable Architecture** designed to support future AI-based automation and intelligent decision-making.

Business Vision

The current prototype demonstrates the feasibility of an affordable smart home automation system. In the future, The Nexus Dome can be developed into a commercial product by offering complete smart home installation and maintenance services.

The business model can include:

- Smart home hardware installation for residential buildings.
- Subscription-based cloud monitoring and remote access services.
- Annual maintenance and technical support.
- Premium automation packages with additional smart devices.
- AI-powered features and analytics available through future software updates.

The modular design allows customers to start with a basic setup and gradually upgrade their homes according to their requirements and budget.

Prototype Cost Analysis

The complete prototype was developed with an approximate hardware cost of **₹3,000**, demonstrating that an effective IoT-based smart home system can be built using affordable and easily available components.

This low-cost approach makes **The Nexus Dome** suitable for:

- Educational demonstrations
- Student research projects
- Small residential homes
- Future commercial expansion

Why Choose The Nexus Dome?

Compared to conventional smart home solutions, The Nexus Dome offers:

- Affordable implementation cost.
 - Integrated monitoring and automation.
 - Remote accessibility through web and mobile platforms.
 - Enhanced home safety using multiple sensors.
 - Modular architecture for future expansion.
 - Easy scalability without replacing the existing system.
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11. Future Scope

Overview

The current implementation of **The Nexus Dome** demonstrates an IoT-based smart home automation ecosystem capable of real-time monitoring, remote control, and rule-based automation. Although the prototype successfully achieves its primary objectives, its modular architecture provides a strong foundation for integrating advanced Artificial Intelligence (AI) and smart technologies in future versions.

The following enhancements are planned to further improve the intelligence, efficiency, and user experience of the system.

1. AI Behaviour Learning

Future versions of **The Nexus Dome** will be capable of learning the daily routines and preferences of users. Instead of relying only on predefined automation rules, the system will analyze historical data and automatically adjust lighting, cooling, and appliance usage according to user behavior.

2. Voice Assistant Integration

The system can be integrated with popular voice assistants such as **Google Assistant** and **Amazon Alexa**, enabling users to control appliances through natural voice commands without manually using the dashboard or mobile application.

3. Computer Vision Based Security

AI-powered computer vision can be incorporated to identify authorized users, recognize unknown visitors, and enhance surveillance. This feature can provide intelligent security monitoring and improve home safety beyond traditional sensor-based detection.

4. Predictive Maintenance

Using machine learning algorithms and historical sensor data, the system can predict potential failures of electrical appliances before they occur. This will help users perform maintenance proactively, reducing equipment downtime and improving reliability.

5. Intelligent Energy Optimization

Future AI models can analyze electricity consumption patterns and recommend optimal appliance usage schedules. This will reduce unnecessary energy consumption and improve overall energy efficiency.

6. Cloud Data Analytics

The cloud platform can be extended to generate detailed analytical reports, including long-term trends of temperature, humidity, power consumption, and security events. These insights will help users better understand and optimize their home's performance.

7. Occupancy Detection

Advanced occupancy detection algorithms can identify whether rooms are occupied and automatically manage lighting, cooling, and other appliances, minimizing unnecessary energy usage.

8. Digital Twin Technology

A digital twin of the house can be developed to create a virtual representation of the physical smart home. This virtual model will enable remote monitoring, simulation, predictive analysis, and easier management of connected devices.

9. Solar & Smart Grid Integration

The Nexus Dome can be integrated with solar energy systems and smart grids to monitor renewable energy generation, optimize electricity consumption, and support sustainable energy management.

Future Vision

The long-term vision of **The Nexus Dome** is to evolve from a rule-based IoT automation system into a fully intelligent smart home ecosystem capable of learning, adapting, and making autonomous decisions. By integrating Artificial Intelligence, cloud computing, and advanced analytics, the platform aims to provide greater convenience, enhanced security, improved energy efficiency, and a better quality of life for users.

12. Conclusion

Overview

The Nexus Dome successfully demonstrates the implementation of an affordable and scalable IoT-based Smart Home Automation Ecosystem. The project integrates environmental monitoring, home security, energy management, and remote accessibility into a unified platform using modern hardware and software technologies.

Throughout the development process, various IoT sensors were integrated with the ESP32 microcontroller to continuously monitor environmental conditions and automate household appliances based on predefined rules. The collected sensor data is securely synchronized with the cloud, allowing users to monitor and control their homes remotely through a responsive web dashboard and Android mobile application.

One of the major strengths of **The Nexus Dome** is its modular architecture, which enables future expansion without requiring significant hardware modifications. This makes the system suitable not only as an educational prototype but also as a foundation for real-world smart home applications.

The prototype was successfully developed using affordable hardware components with an approximate implementation cost of **₹3,000**, demonstrating that advanced smart home

automation can be achieved without relying on expensive proprietary solutions.

Although the current system focuses on IoT-based rule-driven automation, its architecture has been carefully designed to support future Artificial Intelligence integration, including behaviour learning, predictive maintenance, computer vision, and intelligent energy optimization.

Overall, **The Nexus Dome** represents an important step toward developing affordable, intelligent, and connected smart home solutions that improve safety, convenience, and energy efficiency.

13. Project Outcomes

1. The project successfully achieved the following objectives:
 2. Developed a complete IoT-based Smart Home Automation prototype.
 3. Integrated multiple environmental and security sensors with the ESP32.
 4. Implemented real-time monitoring through a responsive web dashboard.
 5. Developed an Android mobile application for remote access.
 6. Automated household appliances using rule-based decision making.
 7. Improved home safety through gas, fire, motion, earthquake, and keypad-based door lock security.
 8. Monitored household energy consumption using the ACS712 current sensor.
 9. Established a scalable architecture capable of supporting future AI-powered enhancements.
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14. References

The following resources were referred to during the design and development of **The Nexus Dome**:

1. ESP32 Official Documentation – Espressif Systems.
2. Arduino IDE & ESP32 Board Package Documentation.
3. Django Official Documentation.
4. Django REST Framework Documentation.
5. PostgreSQL Official Documentation.
6. React Official Documentation.
7. React Native Official Documentation.
8. Firebase Cloud Messaging Documentation.

9. Wokwi Online IoT Simulator Documentation.
 10. GitHub Documentation for Version Control.
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